

OSSS 8

Question:

To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

Solution:

```
ubuntu@ubuntu:~$ nano variable_expansion.c
ubuntu@ubuntu:~$ gcc variable_expansion.c -o variable_expansion
ubuntu@ubuntu:~$ ./variable_expansion
Enter Variable Name (without $): HOME
Expanded Value: /home/ubuntu
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char input[100];
```

```
    printf("Enter Variable Name (without $): ");
```

```
    scanf("%s", input);
```

```
    char *value = getenv(input);
```

```
    if(value != NULL)
```

```
    {
```

```
        printf("Expanded Value: %s\n", value);
```

```
    }
```

```

else

{

    printf("Undefined Variable\n");

}

return 0;

}

```

```

ubuntu@ubuntu:~$ echo $HOME
/home/ubuntu
ubuntu@ubuntu:~$ echo $USER
ubuntu
ubuntu@ubuntu:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
ubuntu@ubuntu:~$ env
SHELL=/bin/bash
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
QT_IM_MODULES=wayland;ibus
PTYXIS_PROFILE=0d9df0a327d8a1d3ce6b85816abc795d
SSH_AUTH_SOCK=/run/user/1000/gcr/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
DBUS_STARTER_BUS_TYPE=session
PWD=/home/ubuntu
XDG_SESSION_DESKTOP=ubuntu
LOGNAME=ubuntu
XDG_SESSION_TYPE=wayland
GPG_AGENT_INFO=/run/user/1000/gnupg/S.gpg-agent:0:1
SYSTEMD_EXEC_PID=2864
XAUTORITY=/run/user/1000/.mutter-Xwaylandauth.OOBjW3
HOME=/home/ubuntu
USERNAME=ubuntu
LANG=C.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:nf=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:
st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31
:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;3
1:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace
=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;3
5:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:
*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.
ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01
;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au
ubuntu@ubuntu:~$ nano nested_variable.c
ubuntu@ubuntu:~$ gcc nested_variable.c -o nested_variable
ubuntu@ubuntu:~$ ./nested_variable
HOME = /home/ubuntu
USER = ubuntu

```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main()
```

```

{

    printf("HOME = %s\n", getenv("HOME"));

    printf("USER = %s\n", getenv("USER"));

```

```
return 0;
```

```
}
```

```
ubuntu@ubuntu:~$ nano builtin_dispatch.c
ubuntu@ubuntu:~$ gcc builtin_dispatch.c -o builtin_dispatch
ubuntu@ubuntu:~$ ./builtin_dispatch
Enter Command: pwd
Executing Built-in: pwd
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void execute_builtin(char *cmd)
```

```
{
```

```
    if(strcmp(cmd, "pwd") == 0)
```

```
    {
```

```
        printf("Executing Built-in: pwd\n");
```

```
    }
```

```
    else if(strcmp(cmd, "cd") == 0)
```

```
    {
```

```
        printf("Executing Built-in: cd\n");
```

```
    }
```

```
    else if(strcmp(cmd, "exit") == 0)
```

```
    {
```

```
        printf("Executing Built-in: exit\n");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Invalid Command\n");
```

```
    }
```

```
}
```

```
int main()
```

```
{  
    char cmd[50];  
  
    printf("Enter Command: ");  
    scanf("%s", cmd);  
  
    execute_builtin(cmd);  
  
    return 0;  
}
```